Yanyu Chen

Robotics | Learned Control · Mechanical · Embedded

Contact Information

Telephone：+65 9197 2335 / +86 132 6221 1293

Email：yc.yanyuchen@gmail.com

LinkedIn：https://www.linkedin.com/in/yanyu-c/

Location：Singapore

GitHub: github.com/yammmyu

Summary

First-year NUS Robotics and Machine Intelligence undergraduate. As sole engineer at MUJIN, built and deployed an imitation-learned dual-arm manipulation stack on a humanoid robot, covering teleoperation, data collection, real-time control and safety. Also designs mechanical systems (SolidWorks CSWP) and embedded firmware for NUS Calibur's RoboMaster team. Working proficiency in English, Chinese and German.

Education

National University of Singapore (NUS)
Robotics and Machine Intelligence (B.Eng)
Minor: Innovation and Design | Aug 2025 – May 2029 (expected) | Singapore

Frankfurt International School
International Baccalaureate (IB Diploma)
Aug 2019 – May 2025 | Germany | Score: 43/45 (Top 2% Global) | Honours: 2023-2025 highest distinction, Physics Award (Class of 2025)

Technical Skills

Learned Control & Robotics SW |PyTorch, imitation learning (diffusion policy), Pinocchio, PyBullet, URDF, YOLO/Ultralytics, OpenCV, pytest

Mechanical Design |SolidWorks, Fusion 360, FEA, DFM/DFA; 5+ years CAD and engineering drawing

Embedded Systems|STM32 HAL/LL, Jetson Orin (CUDA), PID, motor control, CAN/RS485

Robotics Related |ROS2, Gazebo, SLAM navigation, LiDAR/vision integration

Coding and Tools |Python, C++, CMake, Git, STM32CubeIDE, Keil

Experiences

MUJIN Corp.

Software Intern, Humanoid Robotics R&D | May – Jul 2026 | Guangzhou, China

Sole engineer on a 10-week proof of concept: imitation-learned dual-arm parcel handling on an AgiBot Genie G1 humanoid.

Full report, code and engineering documentation: https://github.com/yammmyu/MUJIN_Internship_Summer2026

- Delivered an autonomous grasp → lift → label-check → flip → place cycle with self-recovery from failed grasps, running end to end from the robot's own three cameras. **~90% task success** at **~12 s median cycle** over 38 scored trials.
- Built the layer between a **10 Hz** diffusion policy and a **120 Hz** arm controller: time-uniform substep expansion, cross-chunk temporal ensembling and Gaussian trajectory smoothing. Replaced clock-based synchronisation with **absolute row IDs** keyed to the robot's own execution clock, eliminating the motion oscillation caused by variable inference latency.
- Replaced the vendor's **closed-source IK** with a transparent Pinocchio + URDF solver, calibrated to sub-millimetre FK agreement and validated to **<8 mm / 0.03 rad** max deviation across **400+ points**. Running the same solver in simulation and on hardware made pre-execution collision validation possible for the first time.

- Designed a layered safety envelope of **11 named invariants** (latched E-stop, per-command displacement watchdog, workspace bounds, mandatory sim validation before any command reaches the robot), enforced by a **pytest suite run as a launch pre-flight** that blocks startup on regression.
- Isolated the vendor SDK and ROS behind a single boundary so kinematics, simulation, evaluation and the full test suite run on a laptop with no robot access, **roughly halving iteration time**.
- Owned the handover: **bilingual (EN/CN) engineering documentation**, a PoC report stating measured limitations, and offline evaluation tooling (Wilson-CI success scoring, smoothness analysis, sim-based policy evaluation).

NUS Calibur Robotics Competition Team (DJI RoboMaster)

Mechanical & Firmware | Aug 2025 – Present | Singapore

[Mechanical] "Engineer" robot 6-DOF arm design

- Adopted **truss topology** optimization and interlocking design, **reducing fasteners by 40%** and **reduced weight by 25%** compared to the previous generation
- Designed **internal cable routing** system and integrated conductive slip rings, enabling 360° continuous rotation of the end effector
- Applied DFA (Design for Assembly) principles through 4-5 mechanical iterations, combined 3D printing with CNC machining; the overall robotic arm features **quick-release structures** allowing any component to be fully replaced **within 3 min**

[Firmware] Embedded Platform-based Custom Controller

- Full-stack development of a custom controller **(STM32 platform)**
- Deployed Jetson Orin NX **edge computing platform**, configured and debugged sentry aimbot system

Projects

Ybot - Desktop 6-DOF Robotic Arm

Personal Project | Jun 2023 – Aug 2025 | Mechanical + Firmware

- [Mechanical Design] Mechanical structural design in Fusion 360, achieved **30% weight reduction** through **FEA optimisation**. Exported **URDF** to validate the degrees of freedom in simulation.
- [Hardware Architecture] Used **Raspberry Pi** as host computer, drove stepper motors via Arduino boards through CAN bus, completed distributed control system hardware setup

TurtleBot3 Simulated Warehouse Transportation Project (Mechanical / Firmware)

Coursework (CDE2310) | Jan – Apr 2026 | NUS

- Integrated **LiDAR** and vision to achieve SLAM mapping and path planning, completed static and dynamic station delivery tasks; added ArUco-based automatic docking for zero-angle alignment

Max Planck Institute of Biophysics | Gerhard Hummer Lab

Student Researcher | Jun – Jul 2024 | Frankfurt, Germany

- Completed protein molecular dynamics simulations on HPC cluster using GROMACS engine, developed the **Martini 3** coarse-grained modelling workflow, with Python analysis scripts for statistical processing and visualisation of large-scale simulation data

Awards & Certifications

2026 RoboMaster University League (Chongqing) | Engineer Challenge | 2nd Prize / 1st Prize [CONFIRM]

HIMCM Mathematical Modeling Competition | M Award (Top 4 in Europe, 2023)

SolidWorks CSWP (Full Score)

Language

English: Fluent (academic / professional)

Chinese: Native

German: Proficient (TELC C1)